

Data model

Thirty-three tables across counterparty control, the deal lifecycle, the advisory layer and currency risk.

Revision A covered counterparty control and a minimal deal record. This revision covers all four subject areas of the process design, so the model is complete against the diagrams rather than against phase one of the build.

Design and build are separated deliberately. Section 4 marks every table with the phase that needs it, so a developer can build eighteen tables first and the model still holds when the rest arrive.

Companion documents: 2, application programming interface. 3, user interface. 4, architecture.

1. What changed from revision A

Change	Reason
Eighteen tables added	Confirmation and matching, amendments, accrual and journals, settlement, the advisory harness, currency exposure and hedging, and the bank statement line. Section 2 lists them by subject area.
policy_config becomes policy_version	It is now versioned with effective_from and superseded_at, and both deal and check_run record which version they were tested against. Limits were already versioned. The policy that produced them was not, which made a check reproducible only until somebody changed a threshold.
Accrual is stored, not computed on read	Correct for exposure, insufficient for accounting. The ledger needs what was recognised on each day, not a recomputation of what it should have been. Exposure still reads from the stored accrual rather than recomputing.
Confirmation is a first class table, not a column on the deal	The whole value of matching is that the two records arrive by different routes. Folding the confirmation into the deal leaves nothing to match against, and the control disappears.
Money gains a companion representation	Sterling amounts stay integer pence. Foreign currency amounts are integer minor units with an explicit currency column, because the euro obligations in the currency register are not pence and must never be added to anything that is.
counterparty exposure and currency exposure are permanently separated	Different tables, different names, different units, and a stated rule that no query joins them. Section 9.4 sets out why.

2. The model at a glance

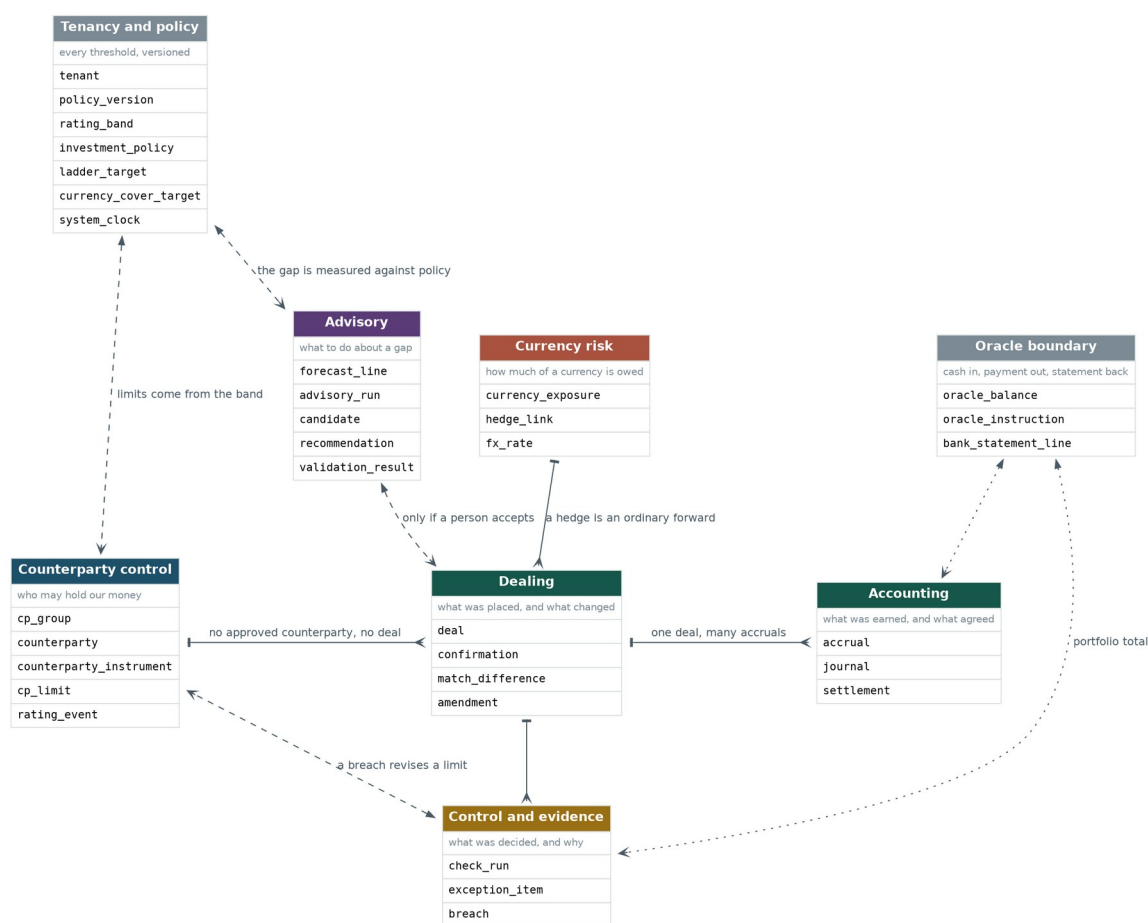


Figure 1. Thirty-three tables in eight subject areas, and how the areas depend on each other.

Four things this map is meant to settle. Counterparty control and dealing are not two systems, they share one database and meet at the limit a deal is tested against and the exposure it then creates. Accounting hangs off dealing rather than sitting beside it. The advisory layer reads policy and writes nothing to the book, so its only path into a deal runs through a person and then through the six checks. And currency risk touches dealing only because a hedge is an ordinary forward, which is why it needs no dealing tables of its own.

2.1 Tables by subject area and build phase

Subject area	Tables	Phase
Tenancy and policy	tenant, policy_version, rating_band, system_clock	1
	investment_policy, ladder_target, currency_cover_target	2
Counterparty control	cp_group, counterparty, counterparty_instrument, cp_limit, rating_event	1
Dealing	deal	1
	confirmation, match_difference, amendment	3
Accounting and settlement	accrual, journal, settlement	2 for accrual and journal, 3 for settlement
Control and evidence	check_run, exception_item, breach	1
Advisory	forecast_line, advisory_run, candidate, recommendation, validation_result	2

Subject area	Tables	Phase
Currency risk	currency_exposure, hedge_link, fx_rate	3
Oracle boundary	oracle_balance, oracle_instruction	1
	bank_statement_line	3

Eighteen tables in phase one, twenty-six by the end of phase two, thirty-three by the end of phase three. The phase one set is unchanged from revision A apart from the policy versioning, so nothing already designed has to be rebuilt.

3. Conventions

3.1 Representation

Concern	Representation	Reason
Sterling money	Integer pence. Suffix _pence.	Never floating point. A rounding drift on a limit check is the one defect a treasury audience will not forgive.
Foreign currency money	Integer minor units. Suffix _minor, with a currency column alongside it, always.	A euro obligation is not pence. Giving it a different suffix makes an accidental addition to a pence column visible in the code rather than only in the result.
Rates and percentages	Integer basis points. Suffix _bp.	4.20 per cent is 420. A 30 per cent cap is 3000.
Exchange rates	Integer scaled by one hundred million. Suffix _e8.	Basis points are too coarse for a rate. Eight decimal places is the market convention and stays exact in integers.
Business dates	ISO 8601 text, YYYY-MM-DD. Suffix _date.	SQLite has no date type. ISO text sorts and compares correctly as text.
Event timestamps	ISO 8601 text. Suffix _at.	Date resolution in this build. Section 13 covers the change under PostgreSQL.
Identifiers	Text, prefixed, random suffix.	Readable on screen and in a log. Stable across a database move. No sequence to reset when reseeded.
Booleans	Integer 0 or 1, with a check constraint.	SQLite has no boolean type.
Structured payloads	JSON in a text column. Only results_json, alternatives_json and payload.	All three are evidence, written once and read whole, never queried by content.

3.2 Rules that hold across every table

Rule	Detail
Nothing is deleted	Superseding a limit, resolving an exception, responding to a breach, reversing an accrual and unlinking a hedge are all writes. There is no delete path and no soft delete flag.
Every operational table carries tenant_id	One seeded value in this build, never queried differently.
A null has exactly one meaning per column	Stated in the column table. Where a null is significant, for example superseded_at meaning current, it is called out.
Nothing computed is stored, with one deliberate exception	Exposure, headroom, coverage and the portfolio total are all derived. Accrual is stored, because the accounting record of what was recognised on a day is a fact rather than a calculation. Section 9 sets out both.
A correction is an entry, not an edit	An amended deal produces reversing accruals and reversing journals. The original rows stay, marked with what reversed them.

4. Entity relationship diagrams

Three detailed views, one per new subject area. Counterparty control is unchanged from revision A apart from the policy versioning and appears in section 5.

4.1 Dealing, accounting and settlement

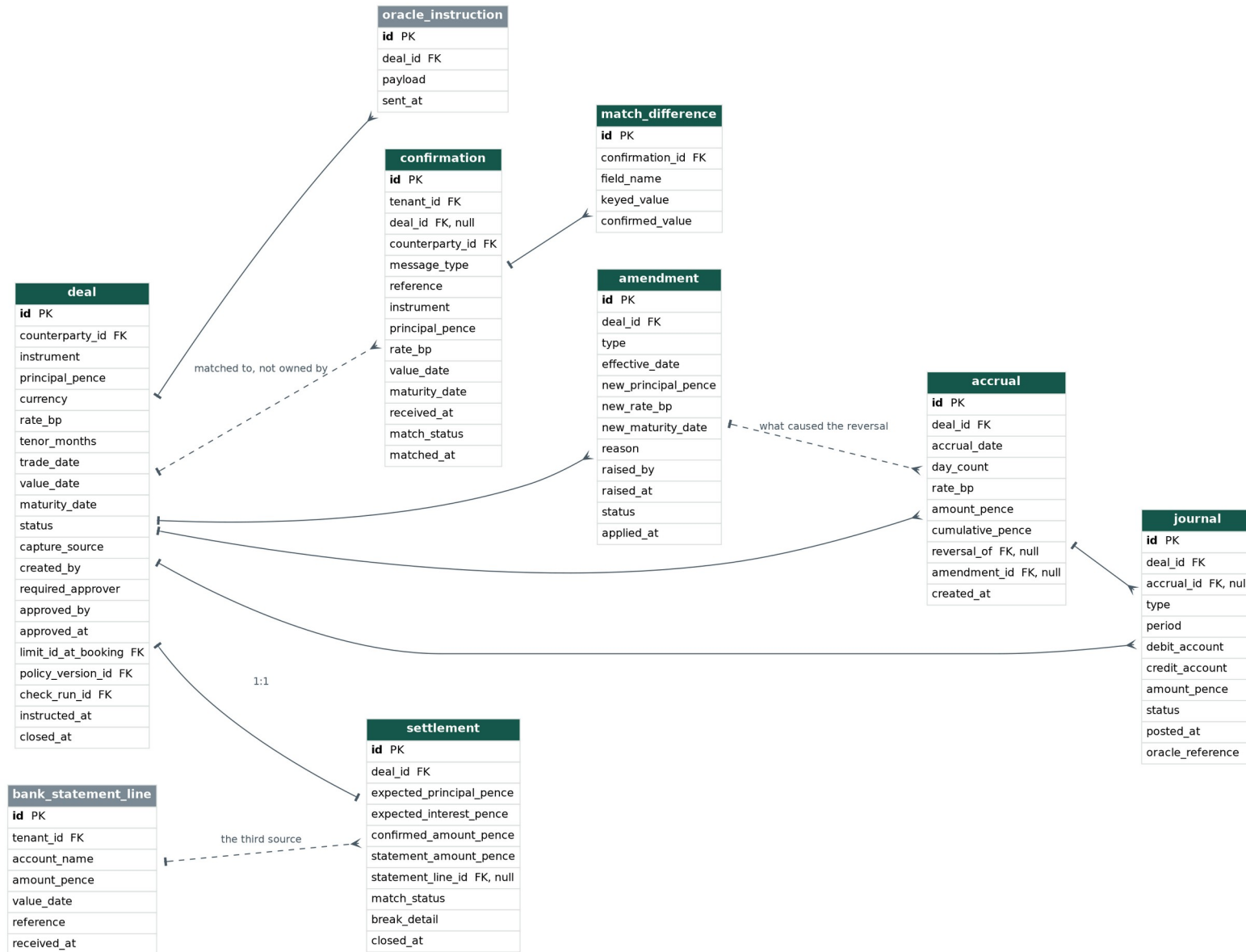


Figure 2. Model 2 of the process design, in full.

The deal is the spine. A confirmation arrives independently and is matched to it rather than owned by it, which is the only reason matching is a control. An amendment causes reversing accruals, and those reversals carry the amendment that caused them. Settlement is where three sources have to agree, and two of three agreeing is not enough.

4.2 The advisory layer

Read the direction of the arrows. Policy and forecast flow in. Candidates and a recommendation flow out. The only edge into the deal table is dashed and labelled, because nothing is recorded because a model suggested it.

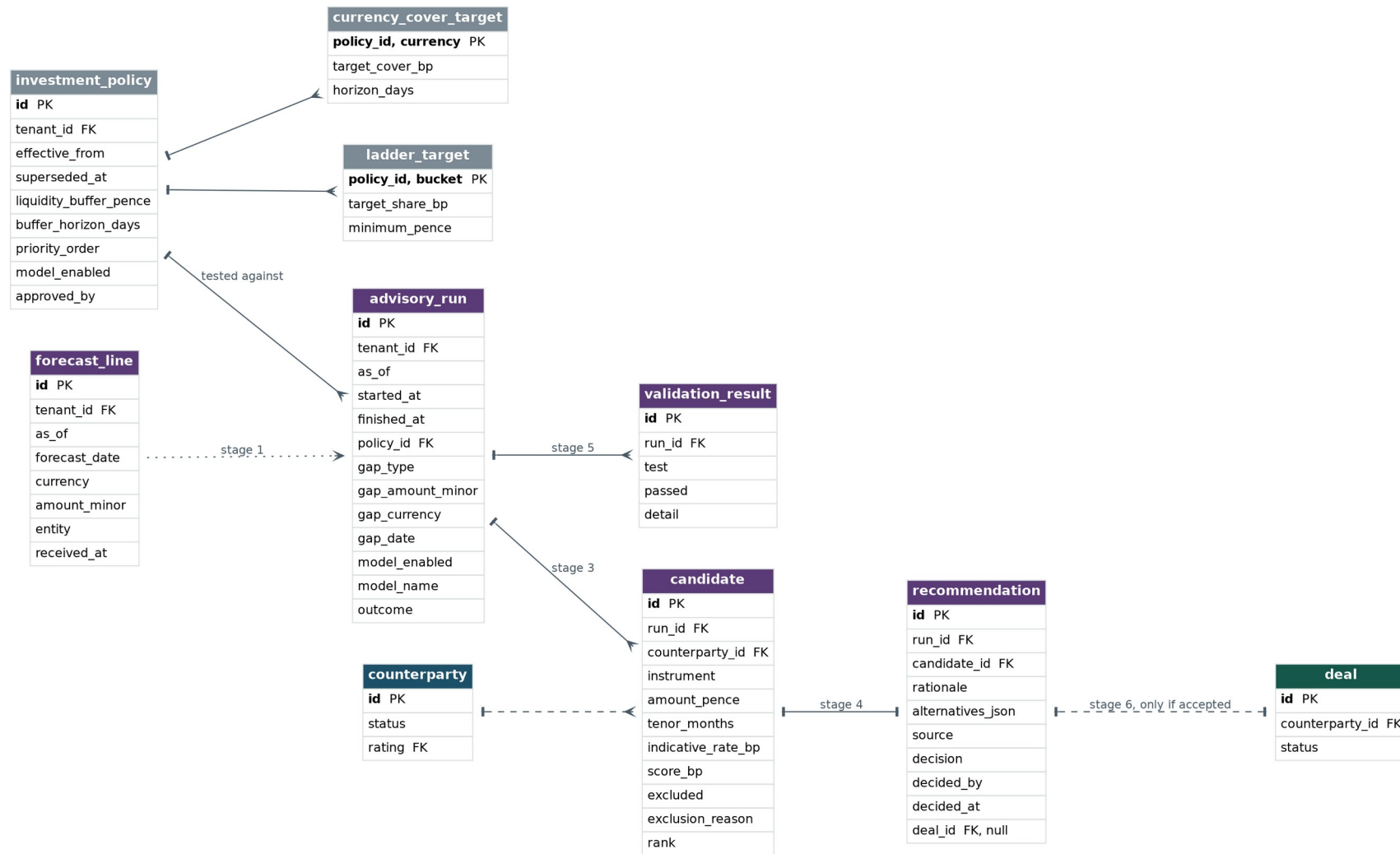


Figure 3. Model 3 of the process design. One nightly run, and the evidence it leaves.

4.3 Currency risk

Three tables. An obligation, the forwards linked to it, and the rates used to value them. The forward itself is an ordinary row in the deal table, which is why there is no separate hedge table.

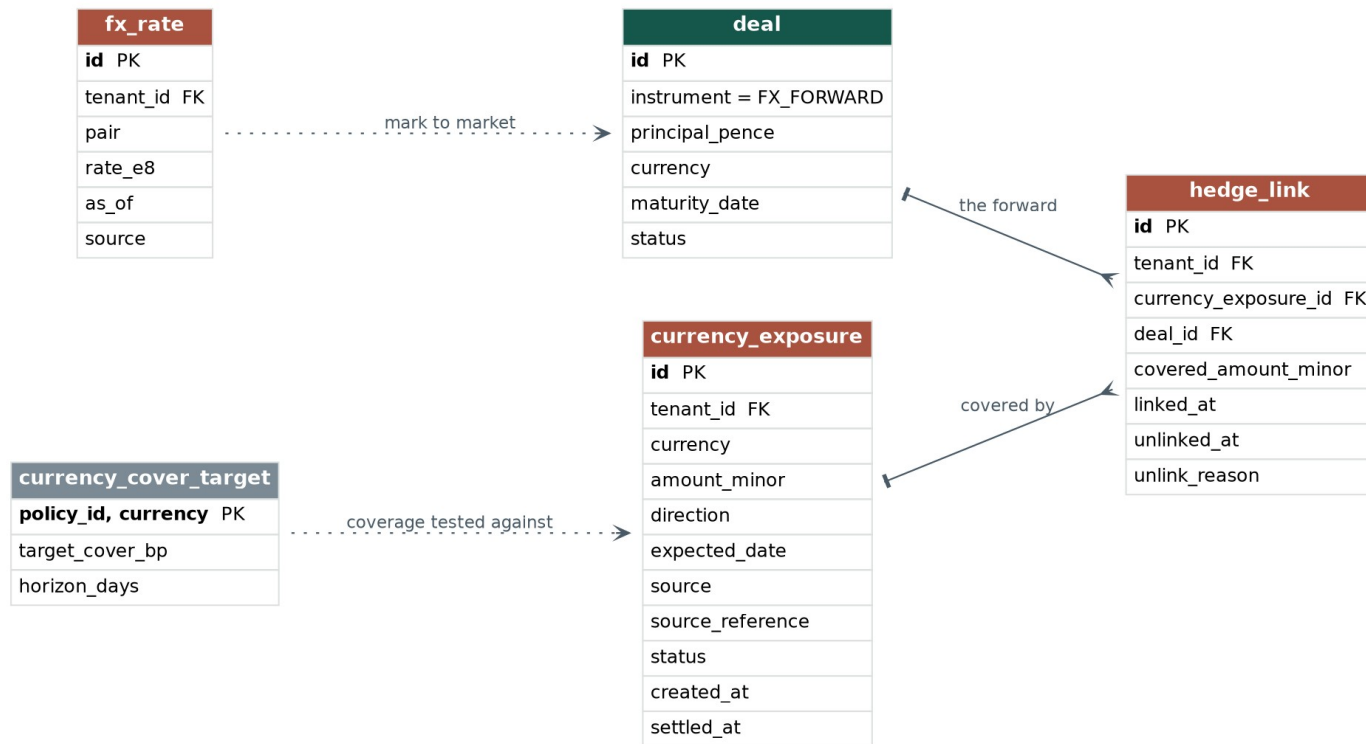


Figure 4. Model 4 of the process design. Only the exposure side is new.

Unchanged from revision A except for policy_version. Reproduced here so the document stands alone.



The treasury policy in force. Versioned, so a check made last quarter can be reproduced against the rules that applied then.

This is the change that closes the reproducibility gap in revision A. A check run now records both the limit version and the policy version, so every input to a historic decision is recoverable rather than only the ones that happened not to change.

Treasury Register · Data model · Document 1 of 4 · Revision B

6. Entity catalogue, dealing

deal

What was placed, with whom, on what terms, and under which limit and policy version. Two columns are new in this revision.

Column	Type	Null	Description
id	text	no	Primary key.
tenant_id, counterparty_id	text	no	Foreign keys.
instrument	text	no	DEPOSIT, FX_FORWARD, MMF or GILT.
principal_pence	integer	no	Face amount. Never the exposure figure.
currency	text	no	The currency of the principal. Sterling for a deposit. For a forward, the currency bought.
rate_bp	integer	no	The contracted rate, or the forward rate.
tenor_months	integer	no	Stored rather than derived, because the tenor check reads it directly.
trade_date, value_date, maturity_date	text	no, no, yes	Maturity is null for open ended holdings.
status	text	no	PROPOSED, BLOCKED, ACTIVE, MATURED, CLOSED or CANCELLED. MATURED is new, and separates a position that is due from one that has been reconciled and closed.
capture_source	text	no	NEW. KEYED or FROM_CONFIRMATION. Auto creating from the confirmation removes the keying error rather than catching it later, and which route was used is worth keeping.
created_by, required_approver, approved_by, approved_at	text	no, yes, yes, yes	
limit_id_at_booking	text	yes	Which limit version the deal passed against.
policy_version_id	text	yes	NEW. Which policy version the checks read.
check_run_id	text	yes	The evidence of the gate decision.
instructed_at, closed_at	text	yes	closed_at is new, set by settlement.

confirmation

The counterparty's own record of what was agreed, arriving by a different route from the deal. Without it you only have your own word for what was traded.

Column	Type	Null	Description
id	text	no	Primary key.
tenant_id	text	no	
deal_id	text	yes	Null until matched. A confirmation can arrive before the deal is keyed, or for a deal that was never keyed at all, and both cases are the point.
counterparty_id	text	yes	Resolved from the message. Null when it cannot be.
message_type	text	no	MT300, MT320, MT535, MT536, BROKER_NOTE or DOCUMENT.
reference	text	no	The bank's own reference.

Column	Type	Null	Description
instrument	text	no	As stated by the counterparty.
principal_pence, rate_bp, value_date, maturity_date	mixed	no	The confirmed terms, held separately from the keyed terms so the two can be compared rather than merged.
received_at	text	no	When it arrived.
match_status	text	no	UNMATCHED, MATCHED, MISMATCHED or DISPUTED.
matched_at	text	yes	
raw_payload	text	yes	The original message, kept whole. Never parsed twice.

match_difference

What disagreed, field by field. One row per field, so the interface can say the rate was keyed at 4.30 and confirmed at 4.28 rather than reporting a mismatch.

Column	Type	Null	Description
id	text	no	Primary key.
confirmation_id	text	no	Foreign key.
field_name	text	no	principal_pence, rate_bp, value_date, maturity_date or instrument.
keyed_value	text	no	As text, because the fields have different types and this table exists to be displayed rather than computed on.
confirmed_value	text	no	

amendment

A break, roll, partial drawdown or correction. The second and last time a person touches a deal.

Column	Type	Null	Description
id	text	no	Primary key.
deal_id	text	no	Foreign key.
type	text	no	BREAK, ROLL, PARTIAL_DRAWDOWN or CORRECTION.
effective_date	text	no	The date from which the new terms apply. Everything recognised after this date has to be recalculated.
new_principal_pence, new_rate_bp, new_maturity_date	mixed	yes	Only the terms that changed. A null means unchanged.
reason	text	no	Why. Required, because an amendment that reverses posted journals needs an explanation.
raised_by, raised_at	text	no	
status	text	no	RAISED, APPLIED or REJECTED.
applied_at	text	yes	When the recalculation and the reversals completed.

Recalculating is the easy half. Knowing what was already recognised is the hard half, and that is only possible because accrual is stored per day rather than computed on read.

7. Entity catalogue, accounting and settlement

accrual

What was recognised, per deal, per day. Stored rather than recomputed.

Column	Type	Null	Description
id	text	no	Primary key.
tenant_id, deal_id	text	no	Foreign keys.
accrual_date	text	no	The day this amount was earned. One row per deal per day.
day_count	integer	no	Usually one. More after a gap, and the column makes the gap visible.
rate_bp	integer	no	The rate used on that day. An amendment changes it from the effective date onward, and the old rows keep the old rate.
amount_pence	integer	no	Signed. Negative on a reversal.
cumulative_pence	integer	no	Running total to this date. Denormalised on purpose, because the alternative is a window function on every read of the blotter.
reversal_of	text	yes	The accrual row this one reverses. Null on an original.
amendment_id	text	yes	What caused the reversal.
created_at	text	no	

journal

What the accounting entry should be. The platform works it out. Oracle posts and keeps it.

Column	Type	Null	Description
id	text	no	Primary key.
tenant_id, deal_id	text	no	Foreign keys.
accrual_id	text	yes	Null for a settlement or revaluation entry.
type	text	no	ACCRUAL, SETTLEMENT, REVALUATION or REVERSAL.
period	text	no	The accounting period, as YYYY-MM. Whether a closed period can be reopened is an accounting policy decision rather than a technical one, and this column is where it bites.
debit_account, credit_account	text	no	From the mapping. Deposit interest, an FX gain and a gilt coupon go to different accounts, and Oracle cannot hold that mapping because the deal does not exist there.
amount_pence	integer	no	Signed.
status	text	no	BUILT, POSTED or FAILED.
posted_at	text	yes	
oracle_reference	text	yes	What Fusion returned. The only evidence that the entry landed.

settlement

Maturity and close. Where three sources have to agree.

Column	Type	Null	Description
id	text	no	Primary key.
tenant_id, deal_id	text	no	One settlement per deal.
expected_principal_pence, expected_interest_pence	integer	no	What the deal record says should arrive.
confirmed_amount_pence	integer	yes	What the counterparty's confirmation says.
statement_amount_pence	integer	yes	What the bank statement shows.

Column	Type	Null	Description
statement_line_id	text	yes	The statement line it matched to.
match_status	text	no	PENDING, AGREED or BREAK.
break_detail	text	yes	Which two of the three agree, and by how much the third differs. Two of three agreeing is not enough: if the record and the confirmation agree but the statement differs, the money did not arrive as promised.
closed_at	text	yes	When the deal was closed and the headroom released.

bank_statement_line

The third source. Supplied by Oracle, prior day rather than intraday.

Column	Type	Null	Description
id	text	no	Primary key.
tenant_id	text	no	
account_name	text	no	
amount_pence	integer	no	Signed. Negative on a payment out.
value_date	text	no	
reference	text	yes	What the bank quoted. Usually, but not always, the deal reference.
received_at	text	no	

8. Entity catalogue, the advisory layer

Eight tables. Three hold the policy the gap is measured against, one holds the forecast, and four hold what a run produced. Nothing here writes to a deal.

investment_policy

The buffer, the ladder and the cover targets. Versioned, like the treasury policy.

Column	Type	Null	Description
id	text	no	Primary key.
tenant_id	text	no	
effective_from, superseded_at	text	no, yes	Null superseded_at means current. One current version per tenant.
liquidity_buffer_pence	integer	no	Cash that must remain available.
buffer_horizon_days	integer	no	Over what period the buffer must hold.
priority_order	text	no	The order of security, liquidity and yield. This is what the ranking stage is told to optimise for, and it is data rather than a prompt.
model_enabled	integer	no	0 or 1. Switching it off drops ranking to a weighted score. The recommendation survives, only the explanation is lost.
approved_by	text	no	

The process design flags that this policy may not exist at the customer yet. If so, the absence is a blocker for the advisory layer and an opportunity to shape the policy, and the table is where that conversation lands.

ladder_target

What share of the portfolio should mature in each bucket.

Column	Type	Null	Description
policy_id	text	no	Primary key, part one.
bucket	text	no	Primary key, part two. 0_3M, 3_6M, 6_12M or OVER_12M.
target_share_bp	integer	no	The target share of the portfolio for this bucket.
minimum_pence	integer	no	An absolute floor, for when the portfolio is small enough that a share is meaningless.

currency_cover_target

How much of a currency obligation must be hedged, and how far out.

Column	Type	Null	Description
policy_id	text	no	Primary key, part one.
currency	text	no	Primary key, part two.
target_cover_bp	integer	no	Required coverage. 8000 is 80 per cent.
horizon_days	integer	no	How far ahead the target applies.

forecast_line

The forecast from Oracle EPM. Interface I-9. Without it the advisory layer has nothing to test.

Column	Type	Null	Description
id	text	no	Primary key.
tenant_id	text	no	
as_of	text	no	The date the forecast was produced. A forecast is a statement made on a day and it is not corrected in place.
forecast_date	text	no	The day the money is expected to move.
currency	text	no	
amount_minor	integer	no	Signed. Positive on an inflow. Minor units of the stated currency.
entity	text	yes	Which part of the organisation.
received_at	text	no	

advisory_run

One nightly run. Written whether or not a model was called, and whether or not a gap was found.

Column	Type	Null	Description
id	text	no	Primary key.
tenant_id	text	no	
as_of	text	no	The clock date the run read.
started_at, finished_at	text	no, yes	
policy_id	text	no	Which investment policy version the gap was measured against.
gap_type	text	no	CASH_SURPLUS, CASH_SHORTFALL, LADDER_GAP, CURRENCY_UNCOVERED or NONE. With NONE the model is never called.
gap_amount_minor, gap_currency, gap_date	mixed	yes	The gap. Null when the type is NONE.
model_enabled	integer	no	As it stood at the time of the run, not as it stands now.
model_name	text	yes	Which model was called. Null when it was not.

Column	Type	Null	Description
outcome	text	no	NO_GAP, MODEL_ACCEPTED, MODEL_REJECTED_FALLBACK or RULE_ONLY.

candidate

The options the rules produced, priced. Excluded options are kept, because why was X not picked is the question that gets asked.

Column	Type	Null	Description
id	text	no	Primary key.
run_id	text	no	Foreign key.
counterparty_id	text	no	Foreign key.
instrument	text	no	
amount_pence	integer	no	
tenor_months	integer	no	
indicative_rate_bp	integer	no	Computed at this stage. The model never generates a rate, and this column is where every figure in the recommendation comes from.
score_bp	integer	no	The deterministic score, computed whether or not the model runs, so the fallback is always available.
excluded	integer	no	0 or 1. Excluded candidates never reach the model.
exclusion_reason	text	yes	Why. For example, three months is its ceiling at BBB+ and the gap runs longer.
rank	integer	yes	Where the model or the score placed it.

recommendation

What was proposed and what a person did about it.

Column	Type	Null	Description
id	text	no	Primary key.
run_id	text	no	Foreign key.
candidate_id	text	no	The chosen candidate. A foreign key, which is what makes an invented counterparty structurally impossible.
rationale	text	yes	The model's prose. Null when the fallback produced the pick.
alternatives_json	text	no	The candidates not picked, with their reasons, kept as shown.
source	text	no	MODEL or RULE_FALLBACK. Shown to the user, not hidden.
decision	text	yes	ACCEPTED, EDITED, REJECTED or EXPIRED. Null while outstanding.
decided_by, decided_at	text	yes	
deal_id	text	yes	Set only if it became a deal, and only after that deal passed the six checks.

The disagreement log is the most valuable data this produces. Every rejection with its reason is a record of where the ranking disagreed with a treasurer, and that is worth more than the ranking itself.

validation_result

Three tests, all of which must pass. One row each, so a failure names itself.

Column	Type	Null	Description
id	text	no	Primary key.

Column	Type	Null	Description
run_id	text	no	Foreign key.
test	text	no	ID_IS_REAL, CHECKS_RERUN_CLEAN or FIGURES_AGREE.
passed	integer	no	0 or 1.
detail	text	yes	What failed, where it failed.

9. Entity catalogue, currency risk

currency_exposure

A future obligation in another currency. It exists before any hedge and often outlives several of them.

Column	Type	Null	Description
id	text	no	Primary key.
tenant_id	text	no	
currency	text	no	Always present. There is no default currency on this table.
amount_minor	integer	no	Minor units of that currency. Not pence, and the suffix says so.
direction	text	no	PAYABLE or RECEIVABLE. Inflows offset outflows, and hedging the gross is buying cover you do not need.
expected_date	text	no	When the money is expected to move.
source	text	no	CONTRACT, PURCHASE_ORDER, INVOICE or FORECAST.
source_reference	text	yes	The purchase order or contract number.
status	text	no	IDENTIFIED, PARTIALLY_COVERED, COVERED or SETTLED.
created_at, settled_at	text	no, yes	

hedge_link

What covers what. The step that makes it hedging rather than owning forwards.

Column	Type	Null	Description
id	text	no	Primary key.
tenant_id	text	no	
currency_exposure_id	text	no	Foreign key.
deal_id	text	no	Foreign key to the FX forward. An ordinary deal row.
covered_amount_minor	integer	no	One forward can cover several exposures and one exposure can take several forwards, so the amount lives on the link rather than on either end.
linked_at	text	no	
unlinked_at	text	yes	Null while the link holds.
unlink_reason	text	yes	ROLLED, CLOSED_EARLY, EXPOSURE_CANCELLED or REALLOCATED.

fx_rate

Rates used to value a forward and to convert an obligation.

Column	Type	Null	Description
id	text	no	Primary key.
tenant_id	text	no	

Column	Type	Null	Description
pair	text	no	For example GBPEUR. Always in one direction, so nothing has to guess which way to divide.
rate_e8	integer	no	The rate scaled by one hundred million.
as_of	text	no	One rate per pair per date.
source	text	no	Where it came from. Stubbed in this build.

9.4 Two different things are called exposure

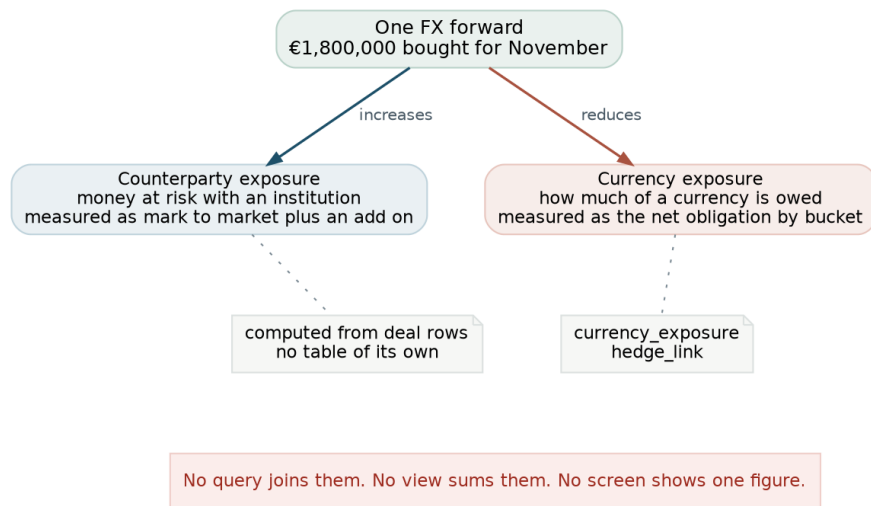


Figure 6. Same word, opposite direction, separate tables.

This is the single most likely modelling mistake in the whole system, and the defence is structural rather than documentary.

Rule	Enforcement
They never share a table	Counterparty exposure has no table. It is computed from deal rows. Currency exposure has its own.
They never share a unit	Counterparty exposure is pence. Currency exposure is minor units with an explicit currency.
They never share a name	No column, view or function is called exposure without a qualifier.
No query joins them	There is no key that would let one. currency_exposure has no counterparty_id, because an obligation to a supplier is not an obligation to a bank.
No screen shows one figure	The interface puts them on separate tabs that cannot be open at once, which document 3 covers.

10. Derived data

Four figures are computed on every read. One is stored, and the exception is deliberate.

Figure	Definition	Stored
Entity exposure	Sum of the measure over active deals for one counterparty.	No
Group exposure	Same, joined to counterparty on group_id.	No
Portfolio total	Same over all active deals, plus the Oracle balance as of the clock date.	No

Figure	Definition	Stored
Currency coverage	Net obligation by currency and time bucket, less the sum of covered_amount_minor on live hedge links, against the policy target.	No
Accrual	Interest earned per deal per day.	Yes

Accrual is the exception because it is not the same kind of thing. Exposure answers what is true now, and recomputing it is the only way to be sure it agrees with the deals behind it. An accrual answers what was recognised on the third of October, which is a historic fact that a later recomputation would quietly rewrite. Once a journal has been posted against it, changing it is a reversal rather than a recalculation.

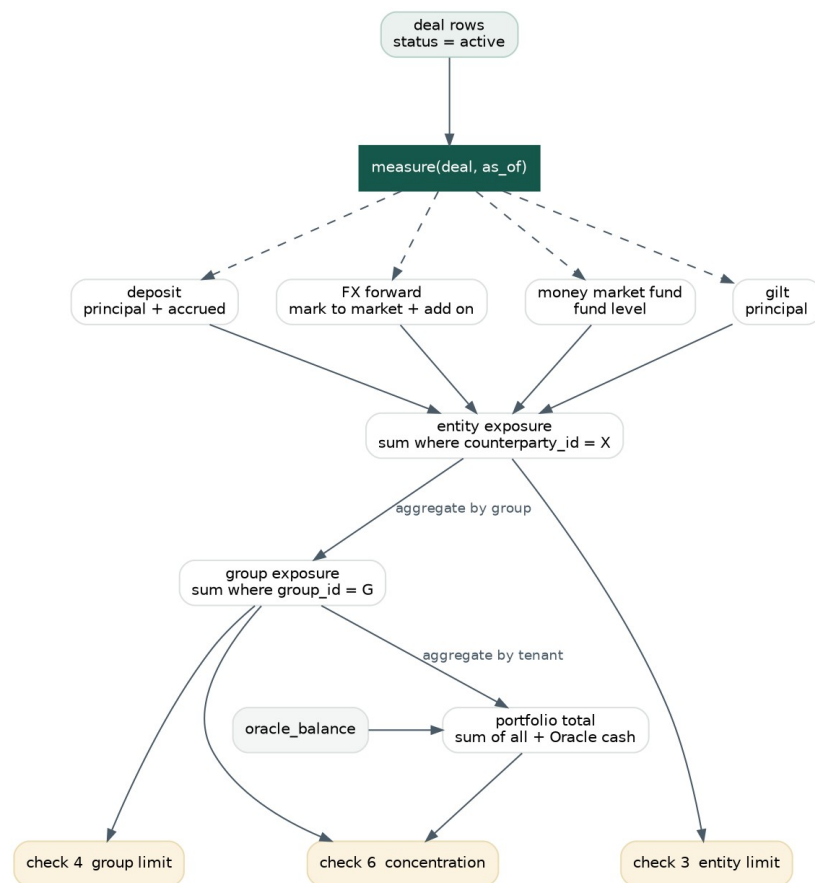


Figure 7. How counterparty exposure is derived, and which checks read which aggregation.

11. Lifecycles

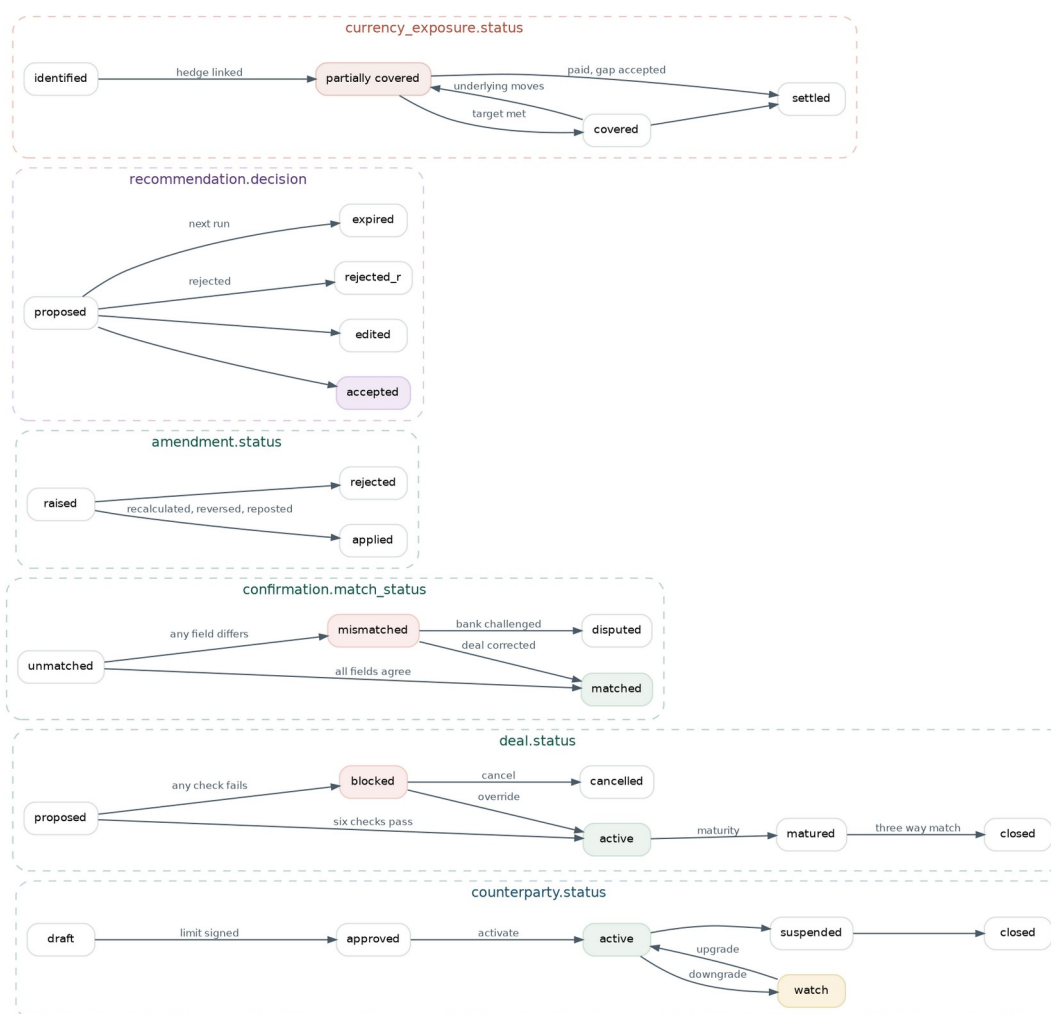


Figure 8. Permitted transitions across all four subject areas. Anything not drawn is not allowed.

Transition	Why it matters
counterparty approved to active	The boundary between the slow onboarding loop and the fast per deal loop. The only thing that puts a name in front of a dealer.
deal blocked to active	The override. The only transition that requires a recorded reason.
deal active to matured to closed	Matured means due. Closed means three sources agreed. Until closed, the counterparty's headroom is still consumed.
confirmation mismatched to matched	Reached by correcting the deal, which reruns the six checks, because a corrected rate changes the accrual and can change the measured exposure.
recommendation proposed to expired	A recommendation not acted on before the next run expires rather than persisting. Yesterday's advice against today's forecast is worse than no advice.
currency_exposure covered to partially covered	Coverage drifts backwards when the underlying moves. Nobody made a mistake. The world moved, and the model has to allow the status to go back.

12. Integrity

12.1 New indexes

Index	On	Exists for
ux_policy_current	policy_version (tenant_id) where superseded_at is null	One current treasury policy per tenant. Partial unique.
ux_investment_policy_current	investment_policy (tenant_id) where superseded_at is null	The same, for the investment policy.
ux_accrual_day	accrual (deal_id, accrual_date) where reversal_of is null	One original accrual per deal per day. Partial unique, so reversals can share a date with what they reverse.
ux_settlement_deal	settlement (deal_id)	One settlement per deal.
ix_confirmation_unmatched	confirmation (tenant_id, match_status) where deal_id is null	The matching queue. Confirmations looking for a deal.
ix_journal_period	journal (tenant_id, period, status)	What is unposted in an open period.
ix_candidate_run	candidate (run_id, excluded, rank)	The candidate list as displayed.
ix_hedge_live	hedge_link (currency_exposure_id) where unlinked_at is null	Live coverage for one exposure.
ix_forecast_window	forecast_line (tenant_id, forecast_date)	The forecast the nightly run reads.

12.2 New invariants the schema cannot hold

Invariant	Why the schema cannot hold it	Where it lives
A confirmation may only match a deal for the same counterparty and instrument	Cross row comparison.	MatchService
A reversing accrual must not exceed what was originally recognised	Cross row arithmetic.	AmendmentService
A journal may not post into a closed period unless policy allows reopening	Depends on an accounting policy setting rather than on the row.	JournalService
Covered amounts on live hedge links must not exceed the exposure they cover	Sum across rows.	HedgeService
A recommendation may only reference a candidate from its own run	Expressible as a composite foreign key, at the cost of composite keys throughout.	Repository layer
A recommendation may only carry a deal_id for a deal that passed the six checks	Cross table state comparison.	AdvisoryService

Invariant	Why the schema cannot hold it	Where it lives
An excluded candidate may never be ranked	Cross column rule that a check constraint could express, but only per row, and the meaningful version is per run.	AdvisoryService

12.3 What is still absent

Absent	Needed for	Note
User and role tables	Segregation of duties, and the rule that the approver is not the proposer	Actors are still recorded as text names. This is now the largest single gap in the model.
Hedge accounting under IFRS 9	Designation, effectiveness testing, a cash flow hedge reserve	Deliberately out. Forwards are marked to market through the income statement. Needs confirming with the accountants rather than assuming.
Instruments beyond four	Facilities, repo, commercial paper	Each needs an exposure measure and an accrual method before its code can be added to the enumeration.
Multi currency deposits	A deposit in a currency other than the base	The currency column exists. The limit arithmetic assumes one currency, and section 3.1 of document 4 covers what changes.
Intraday balances	A concentration check against a live position	Statements are prior day. A stated boundary rather than a defect.